Expanded action - "Cheat sheet"

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1 Action Ansatz

$$S = S_2 + S_3 \tag{1}$$

where

$$S_2 = \int \frac{d^D p}{(2\pi)^D} \Big[\tilde{\phi}^{\mu\nu}(-p)\, K^{\phi\phi}_{\mu\nu\rho\sigma}(p) \tilde{\phi}_{\rho\sigma}(p) + \tilde{\phi}^{\mu\nu}(-p) K^{\phi G}_{\mu\nu\lambda}{}^{\rho\sigma}(p) \tilde{G}^{\lambda}_{\rho\sigma}(p) + \tilde{G}^{\lambda}_{\rho\sigma}(-p) K^{GG\,\mu\nu}_{\lambda}{}_{\rho}{}^{\sigma\tau}(p) \tilde{G}^{\rho}_{\sigma\tau}(p) \Big] \tag{2}$$

$$\begin{aligned}
S_3 &= S_{\phi\phi\phi} + S_{\phi\phi G} + S_{\phi G G} \\
S_{\phi\phi\phi} &= \int \frac{d^D p}{(2\pi)^D} \frac{d^D q}{(2\pi)^D}\, \tilde{\phi}^{\mu\nu}(-p) \tilde{\phi}^{\alpha\beta}(-q) \tilde{\phi}^{\rho\sigma}(p+q) K^{\phi\phi\phi}_{\mu\nu\alpha\beta\rho\sigma}(p,q) \\
S_{\phi\phi G} &= \int \frac{d^D p}{(2\pi)^D} \frac{d^D q}{(2\pi)^D}\, \tilde{\phi}^{\mu\nu}(-p) \tilde{\phi}^{\alpha\beta}(-q) K^{\phi\phi G}_{\mu\nu\alpha\beta\lambda}{}^{\rho\sigma}(p,q) \tilde{G}^{\lambda}_{\rho\sigma}(p+q) \\
S_{\phi G G} &= \int \frac{d^D p}{(2\pi)^D} \frac{d^D q}{(2\pi)^D}\, \tilde{\phi}^{\mu\nu}(-p-q) K^{\phi G G\,\alpha\beta}_{\mu\nu\lambda}{}_{\rho}{}^{\sigma\tau} \tilde{G}^{\lambda}_{\alpha\beta}(p) \tilde{G}^{\rho}_{\sigma\tau}(q)
\end{aligned} \tag{3}$$

2 Gauge transformations

$$\begin{aligned}
\delta\tilde{\phi}^{\mu\nu} = \delta^{(0)}\tilde{\phi}^{\mu\nu} + \delta^{(1)}\tilde{\phi}^{\mu\nu}(p) &= D^{\mu\nu}_{\phi^0\ \alpha}(p)\tilde{\omega}^{\alpha}(p) + \int \frac{d^D d}{(2\pi)^D}\, q\tilde{\phi}^{\alpha\beta}(p-q)\tilde{\omega}^{\rho}(q) D^{\mu\nu}_{\phi^1\ \alpha\beta\rho}(p,q) \\
D^{\mu\nu}_{\phi^0\ \alpha}(p) &= -i(\eta^{\mu\nu}p_\alpha - p^{(\mu}\delta^{\nu)}_{\alpha}) \\
D^{\mu\nu}_{\phi^1\ \alpha\beta\rho}(p,q) &= -i(\delta^{\mu}_{\alpha}\delta^{\nu}_{\beta}p_\rho - q_\beta\delta^{(\mu}_{\alpha}\delta^{\nu)}_{\rho})
\end{aligned} \tag{4}$$

$$\begin{aligned}
\delta\tilde{G}^{\lambda}_{\mu\nu}(p) = \delta^{(0)}\tilde{G}^{\lambda}_{\mu\nu}(p) + \delta^{(1)}\tilde{G}^{\lambda}_{\mu\nu}(p) &= D^{\lambda}_{G^0\ \mu\nu\alpha}(p)\tilde{\omega}^{\alpha}(p) + \int \frac{d^D d}{(2\pi)^D}\, q\tilde{G}^{\alpha}_{\beta\gamma}(p-q)\tilde{\omega}^{\rho}(q) D^{\lambda}_{G^1\ \mu\nu\alpha}{}^{\beta\gamma}{}_{\rho}(p,q) \\
D^{\lambda}_{G^0\ \mu\nu\alpha}(p) &= -p_\mu p_\nu \delta^{\lambda}_{\alpha} + \frac{1}{2}\delta^{\lambda}_{(\mu}p_{\nu)}p_\alpha \\
D^{\lambda}_{G^1\ \mu\nu\alpha}{}^{\beta\gamma}{}_{\rho}(p,q) &= -i[\delta^{\lambda}_{\alpha}\delta^{\beta}_{\mu}\delta^{\gamma}_{\nu}(p-q)_\rho - q_\alpha\delta^{\beta}_{\mu}\delta^{\gamma}_{\nu}\delta^{\lambda}_{\rho} + \delta^{\lambda}_{\alpha}\delta^{\beta}_{\rho}\delta^{\gamma}_{(\mu}q_{\nu)}]
\end{aligned} \tag{5}$$

3 Old Identities

$$\mathcal{W}^{\text{old }\phi}_{\rho\sigma\alpha}(p) \equiv 2D^{\mu\nu}_{\phi^0\ \alpha}(p)\, K^{\phi\phi}_{\mu\nu\rho\sigma}(p) + D^{\lambda}_{G^0\ \mu\nu\alpha}(p)\, K^{\phi G\ \mu\nu}_{\rho\sigma\lambda}(p) = 0 \tag{6}$$

$$\mathcal{W}^{\text{old }G}_{\rho\sigma\tau\alpha}(p) \equiv D^{\mu\nu}_{\phi^0\ \alpha}(p)\, K^{\phi G}_{\mu\nu\rho\sigma\tau}(-p) + 2D^{\lambda}_{G^0\ \mu\nu\alpha}(p)\, K^{GG\ \mu\nu}_{\lambda\ \rho\sigma\tau} = 0 \tag{7}$$

$$\mathcal{W}^{\text{old }31}_{\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \equiv K^{\phi\phi}_{\mu_1\nu_1\alpha\beta}(p_1) D^{\phi^1\alpha\beta}_{\ \ \mu_2\nu_2\rho}(p_1,p_1+p_2) + K^{\phi\phi}_{\mu_2\nu_2\alpha\beta}(p_2) D^{\phi^1\alpha\beta}_{\ \ \mu_1\nu_1\rho}(p_2,p_1+p_2) = 0 \tag{8}$$

$$\begin{aligned}\mathcal{W}^{\text{old }32}_{\lambda_1\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & 2K^{\phi GG}_{\mu_2\nu_2\lambda\mu\nu\lambda_1\mu_1\nu_1} D^{G^0\lambda\mu\nu}_{\ \ \ \ \ \ \rho}(p_1+p_2) + K^{\phi G}_{\mu\nu\lambda_1\mu_1\nu_1}(-p_1) D^{\phi^1\mu\nu}_{\ \ \ \ \ \mu_2\nu_2\rho}(p_1,p_1+p_2) + \\ & + K^{\phi G}_{\mu_2\nu_2\lambda\ \ \ }{}^{\mu\nu}(p_2) D^{G^1\lambda}_{\ \ \ \mu\nu\lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2) = 0\end{aligned} \tag{9}$$

$$\begin{aligned}\mathcal{W}^{\text{old }33}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2} D^{\phi^0\mu\nu}_{\ \ \ \ \ \ \rho}(p_1+p_2) + K^{GG}_{\lambda\mu\nu\lambda_1\mu_1\nu_1} D^{G^1\lambda\mu\nu}_{\ \ \ \ \ \ \lambda_2\mu_2\nu_2\rho}(p_1,p_1+p_2) + \\ & + K^{GG}_{\lambda\mu\nu\lambda_2\mu_2\nu_2} D^{G^1\lambda\mu\nu}_{\ \ \ \ \ \ \lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2) = 0\end{aligned} \tag{10}$$

$$\begin{aligned}\mathcal{W}^{\text{old }34}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\mu_3\nu_3\rho}(p_1,p_2,p_3) \equiv\ & K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_1\mu_1\nu_1} D^{G^1\lambda\mu\nu}_{\ \ \ \ \ \ \lambda_2\mu_2\nu_2\rho}(p_1+p_3,p_1+p_2+p_3) + \\ & + K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_2\mu_2\nu_2} D^{G^1\lambda\mu\nu}_{\ \ \ \ \ \ \lambda_1\mu_1\nu_1\rho}(p_2+p_3,p_1+p_2+p_3) + \\ & + K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2} D^{\phi^1\mu\nu}_{\ \ \ \ \ \ \mu_3\nu_3\rho}(p_1+p_2,p_1+p_2+p_3) = 0\end{aligned} \tag{11}$$

4 New Identities

$$\mathcal{W}^{\phi} = \mathcal{W}^{\text{old }\phi}_{\rho\sigma\alpha}(p) \equiv 2D^{\mu\nu}_{\phi^0\ \alpha}(p)\, K^{\phi\phi}_{\mu\nu\rho\sigma}(p) + D^{\lambda}_{G^0\ \mu\nu\alpha}(p)\, K^{\phi G\ \mu\nu}_{\rho\sigma\lambda}(p) = 0 \tag{12}$$

$$\mathcal{W}^{G} = \mathcal{W}^{\text{old }G}_{\rho\sigma\tau\alpha}(p) \equiv D^{\mu\nu}_{\phi^0\ \alpha}(p)\, K^{\phi G}_{\mu\nu\rho\sigma\tau}(-p) + 2D^{\lambda}_{G^0\ \mu\nu\alpha}(p)\, K^{GG\ \mu\nu}_{\lambda\ \rho\sigma\tau} = 0 \tag{13}$$

$$\begin{aligned}\mathcal{W}^{31}_{\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \equiv\ & 3K^{\phi\phi\phi}_{\mu_1\nu_1\mu_2\nu_2\mu\nu}(p_1,p_2) D^{\phi^0\mu\nu}_{\ \ \ \rho}(p_1+p_2) + K^{\phi\phi G}_{\mu_1\nu_1\mu_2\nu_2\lambda\ \ \ }{}^{\mu\nu} D^{\lambda}_{G^0\ \mu\nu\rho}(p_1+p_2) + \mathcal{W}^{\text{old }31}_{\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2) \\ =\ & 3K^{\phi\phi\phi}_{\mu_1\nu_1\mu_2\nu_2\mu\nu}(p_1,p_2) D^{\phi^0\mu\nu}_{\ \ \ \rho}(p_1+p_2) + K^{\phi\phi G}_{\mu_1\nu_1\mu_2\nu_2\lambda\ \ \ }{}^{\mu\nu} D^{\lambda}_{G^0\ \mu\nu\rho}(p_1+p_2) + \\ & + K^{\phi\phi}_{\mu_1\nu_1\mu\nu}(p_1) D^{\phi^1\mu\nu}_{\ \ \ \mu_2\nu_2\rho}(p_1,p_1+p_2) + K^{\phi\phi}_{\mu_2\nu_2\mu\nu}(p_2) D^{\phi^1\mu\nu}_{\ \ \ \mu_1\nu_1\rho}(p_2,p_1+p_2) = 0\end{aligned} \tag{14}$$

$$\begin{aligned}\mathcal{W}^{32}\lambda_1\mu_1\nu_1\mu_2\nu_2\rho(p_1,p_2) &\equiv 2K^{\phi\phi G}_{\mu\nu\mu_2\nu_2\lambda_1\mu_1\nu_1}D^{\phi^0\mu\nu}{}_{\rho}(p_1+p_2)+\mathcal{W}^{\text{old }32}_{\lambda_1\mu_1\nu_1\mu_2\nu_2\rho}(p_1,p_2)\\ &= 2K^{\phi\phi G}_{\mu\nu\mu_2\nu_2\lambda_1\mu_1\nu_1}D^{\phi^0\mu\nu}{}_{\rho}(p_1+p_2)+2K^{\phi GG}_{\mu_2\nu_2\lambda\mu\nu\lambda_1\mu_1\nu_1}D^{G^0\lambda\mu\nu}{}_{\rho}(p_1+p_2)+\\ &+K^{\phi G}_{\mu\nu\lambda_1\mu_1\nu_1}(-p_1)D^{\phi^1\mu\nu}{}_{\mu_2\nu_2\rho}(p_1,p_1+p_2)+K^{\phi G}_{\mu_2\nu_2\lambda}{}^{\mu\nu}(p_2)D^{G^1\lambda}{}_{\mu\nu\lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2)=0\end{aligned} \tag{15}$$

$$\begin{aligned}\mathcal{W}^{33}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\rho} &\equiv \mathcal{W}^{\text{old }33}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\rho}(p_1,p_2)\\ &= K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}D^{\phi^0\mu\nu}{}_{\rho}(p_1+p_2)+K^{GG}_{\lambda\mu\nu\lambda_1\mu_1\nu_1}D^{G^1\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1,p_1+p_2)+\\ &+K^{GG}_{\lambda\mu\nu\lambda_2\mu_2\nu_2}D^{G^1\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2,p_1+p_2)=0\end{aligned} \tag{16}$$

$$\begin{aligned}\mathcal{W}^{34}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\mu_3\nu_3\rho} &\equiv \mathcal{W}^{\text{old }34}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\mu_3\nu_3\rho}(p_1,p_2,p_3)\\ &= K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_1\mu_1\nu_1}D^{G^1\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1+p_3,p_1+p_2+p_3)+\\ &+K^{\phi GG}_{\mu_3\nu_3\lambda\mu\nu\lambda_2\mu_2\nu_2}D^{G^1\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2+p_3,p_1+p_2+p_3)+\\ &+K^{\phi GG}_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}D^{\phi^1\mu\nu}{}_{\mu_3\nu_3\rho}(p_1+p_2,p_1+p_2+p_3)=0\end{aligned} \tag{17}$$

$$\begin{aligned}\mathcal{W}^{35}_{\mu_1\nu_1\mu_2\nu_2\mu_3\nu_3\rho}(p_1,p_2,p_3) &\equiv K^{\phi\phi\phi}_{\mu_1\nu_1\mu_2\nu_2\mu\nu}(p_1,p_2)D^{\phi^1\mu\nu}{}_{\mu_3\nu_3\rho}(p_1+p_2,p_1+p_2+p_3)+\\ &+K^{\phi\phi\phi}_{\mu_2\nu_2\mu_3\nu_3\mu\nu}(p_2,p_3)D^{\phi^1\mu\nu}{}_{\mu_1\nu_1\rho}(p_2+p_2,p_1+p_2+p_3)+\\ &+K^{\phi\phi\phi}_{\mu_3\nu_3\mu_1\nu_1\mu\nu}(p_3,p_1)D^{\phi^1\mu\nu}{}_{\mu_2\nu_2\rho}(p_3+p_1,p_1+p_2+p_3)=0\end{aligned} \tag{18}$$

$$\begin{aligned}\mathcal{W}^{36}_{\lambda_1\mu_1\nu_1\mu_2\nu_2\mu_3\nu_3}(p_1,p_2,p_3) &\equiv K^{\phi\phi G}_{\mu_2\nu_2\mu_3\nu_3\lambda}{}^{\mu\nu}(p_2,p_3)D^{G^1\lambda}{}_{\mu\nu\lambda_1\mu_1\nu_1}(p_2+p_3,p_1+p_2+p_3)+\\ &+K^{\phi\phi G}_{\mu_2\nu_2\mu\nu\lambda_1\mu_1\nu_1}(p_2,p_1+p_2)D^{\phi^1\mu\nu}{}_{\mu_3\nu_3\rho}(p_1+p_2,p_1+p_2+p_3)+\\ &+K^{\phi\phi G}_{\mu_3\nu_3\mu\nu\lambda_1\mu_1\nu_1}(p_3,p_1+p_3)D^{\phi^1\mu\nu}{}_{\mu_2\nu_2\rho}(p_1+p_3,p_1+p_2+p_3)\end{aligned} \tag{19}$$